

Revision Notes for Class 7 Science Chapter 2 – Exploring Substances: Acidic, Basic, and Neutral

Everywhere around us are substances with different properties – some taste sour, some are bitter, some feel slippery, while others do not seem to have any pronounced taste or feel. Scientists classify substances broadly into three categories: acidic, basic, and neutral. These differences are not just about taste, but about real chemical properties that can be observed and tested in very simple ways. This chapter demonstrates the use of natural and synthetic indicators to identify the nature of various everyday materials.

Testing for Acids, Bases, and Neutrals Using Litmus

To identify whether a sample is acidic, basic, or neutral, science often begins with the litmus test. Litmus, a dye obtained from lichens, is available as blue or red litmus paper. When a substance turns blue litmus red, it signals acidity, while turning red litmus blue indicates a base. If there is no change with either, the substance is neutral. This method is one of the most common tests you'll find in school laboratories.

- **Acidic substances:** Turn blue litmus red (e.g., lemon juice, vinegar, amla juice).
- **Basic substances:** Turn red litmus blue (e.g., baking soda solution, soap solution, lime water).
- **Neutral substances:** Do not change the color of either litmus paper (e.g., tap water, salt solution, sugar solution).

By recording observations using different test samples and noting down which group (A: acidic, B: basic, C: neutral) each sample falls into, students learn how everyday items behave in such tests. For example, fruits like lemons and tamarind fall into acids, while detergents and soaps are typically basic.

Natural Indicators: Red Rose and Turmeric

Nature provides its own indicators. Red rose extract is prepared by soaking rose petals in hot water. When added to an acidic solution, the extract remains red, but in the presence of a base, it turns green. There's no color change in a neutral solution. This allows students to repeat the same inquiry but using natural garden or kitchen materials.

- Red in acid, green in base, and unchanged in neutral substances when using red rose extract.
- Other natural indicators like beetroot, turmeric, purple cabbage, and hibiscus can also be explored for their property to change color with acids and bases.

Turmeric is another popular indicator, especially common in Indian households. Turmeric paper turns a bright red color in the presence of a basic solution, but remains yellow when exposed to acids or neutrals. This forms the basis for fun secret writing experiments where a message is written on turmeric-stained paper using a basic solution and revealed by a spray of turmeric water.

Acids and Bases Found in Foods and Home

Edible acids such as citric acid (lemon), lactic acid (curd), tartaric acid (tamarind), and acetic acid (vinegar) are present in many common foods, often giving them their sour taste. Bases, on the other hand, are found in household cleaning agents and are responsible for their slippery feel—such as baking soda and soap solutions. However, not all substances with a bitter taste are basic; for example, bitter gourd is not a base.

Olfactory Indicators

Not all indicators work by changing color. Some, called olfactory indicators, change their smell in acidic or basic media. Onion is a classic example: its scent changes depending on the substance it is mixed with. This kind of indicator is especially useful for those who are visually impaired, as it allows differentiation based on smell instead of color.

- Saturate small strips of clean cloth or filter paper with onion smell, then test with acids and bases to notice the change in odour.
- Such changes help to distinguish between acidic and basic environments.

Neutralisation and Its Applications

A key chemical reaction in this chapter is neutralisation. If an acidic substance is mixed with a base in suitable amounts, the result is a neutral solution—producing salt, water, and some heat. This principle is useful in various real-life situations: for example, treating an ant bite (formic acid) with baking soda (base), treating acidic soil with lime, or neutralising acidic factory waste before releasing it into water bodies.

- $\text{Acid} + \text{Base} \rightarrow \text{Salt} + \text{Water} + \text{Heat}$
- Everyday applications include treating ant stings, adjusting soil pH for agriculture, and processing industrial waste.

Summary Points

- Acids, bases, and neutrals are present everywhere, from our food to common household substances.
- Litmus paper, red rose extract, turmeric, and onion can be used to identify these substances.
- Indicators show their presence through color or smell changes.
- Neutralisation reactions play important roles in health, the environment, and industry.
- Understanding these properties helps us handle substances more safely and creatively in everyday life.

Practice and Application

Students should practice by testing familiar substances using natural indicators, observing the results, and discussing cases like ant bites, soil treatment, and safe disposal of factory waste. Exploring secret writing using turmeric and bases, and experimenting with natural indicators, makes science both fun and practical.

